Phase 2 Experimental Results and Comparison

Research Replication and Extension: Stabilizing Word-Level Speech Explanations via KernelSHAP

Base paper	Pastor et al. (EACL 2024), *Explaining Speech Classification Models via Word-Level Audio Segments and Paralinguistic Features*
Dataset / model	Fluent Speech Commands (FSC) / superb/wav2vec2-base-superb-ic
Phase 2 extension	WA-SHAP: word-level KernelSHAP with hard-silence coalitions
Primary finding	Under matched word-only features, WA-SHAP explanations are more stable across random seeds than WA-LIME explanations.

Scope and attribution

This report separates published findings from results obtained by our implementation. Values labelled **Published** are reproduced from the original paper and are not claimed as our experimental results. Values labelled **Our implementation** were computed from the project result files.

Research gap and improvement

Our research gap is LIME's sensitivity to random perturbation sampling. This is a documented stability concern for post-hoc explainers. We implement WA-SHAP, assigning each aligned word a Shapley value from marginal contributions across word coalitions. The controlled experiment tests whether it reduces seed sensitivity; small inputs are enumerated exactly when feasible.

Evaluation summary

The report includes (1) the original paper's Table 4 extract, (2) our Phase 1 reproduction, (3) our full-test WA-SHAP results, (4) a controlled WA-LIME versus WA-SHAP seed-stability comparison, and (5) an exploratory noise-masking sensitivity study.

1. Published baseline from the original research paper

The following is a transcribed numerical extract of Table 4 from Pastor et al. The original PDF page number is shown below. These are published results, not our implementation results.

EXTRACT FROM ORIGINAL PAPER - NOT OUR EXPERIMENT

Metric	Method	Action	Object	Location
Comprehensiveness (higher)	WA-L1O	0.623	0.627	0.467
	WA-LIME	0.638	0.663	0.481
Sufficiency (lower)	WA-L1O	0.161	0.086	0.063
	WA-LIME	0.165	0.077	0.054

Figure 1. Transcribed extract from Pastor et al. (2024), PDF page 7, Table 4. The table reports FSC comprehensiveness (higher is better) and sufficiency (lower is better) for WA-L1O, WA-LIME, and random attribution.

Published interpretation: both original word-level explainers exceed the random baseline; WA-LIME is generally stronger in the paper's reported faithfulness comparison. Source: Pastor et al. (2024), PDF page 7, Table 4.

2. Our Phase 1 reproduction: original implementation

These results were obtained by our implementation over the FSC test split (3,793 audio samples). They are separate from the published values on the prior page.

Method	Action Compr.	Action Suff.	Object Compr.	Object Suff.	Location Compr.	Location Suff.
WA-L1O	0.633	0.158	0.638	0.086	0.453	0.064
WA-LIME	0.637	0.176	0.657	0.088	0.468	0.063
Random	0.285	0.438	0.243	0.400	0.174	0.291

Table 1. Our implementation: Phase 1 full-test faithfulness reproduction. Comprehensiveness is better when higher; sufficiency is better when lower.

Interpretation: both LOO and LIME clearly outperform random attribution. LIME has modestly higher comprehensiveness than LOO in all three slots. On sufficiency, neither method dominates every slot; for example, LOO is better for Action (0.158 vs. 0.176).

Supporting implementation checks

The qualitative word-attribution example assigns the expected command words to the expected FSC slots: action words influence Action, object words influence Object, and location words influence Location. The full-test paralinguistic analysis finds background noise and reverberation have the largest average effects, showing that this speech classifier is sensitive to acoustic conditions as well as lexical content.

3. Proposed improvement: WA-SHAP

WA-SHAP represents each WhisperX-aligned word as a binary feature. A coalition keeps its present words and hard-silences absent word spans. KernelSHAP estimates each word's contribution across sampled coalitions; our stability test asks whether it reduces LIME's seed sensitivity. Small coalition spaces are enumerated exactly.

> **Exact enumeration coverage:** 83.34% of the 3,793 FSC utterances had a coalition space that fit within the 100-coalition budget, so their Shapley values were computed exactly and deterministically. The remaining longer utterances used sampled KernelSHAP.

Method	Action Compr.	Action Suff.	Object Compr.	Object Suff.	Location Compr.	Location Suff.
WA-SHAP (our Phase 2)	0.659	0.210	0.673	0.117	0.508	0.086

Table 2. Our implementation: WA-SHAP faithfulness over 3,793 FSC test examples. Higher comprehensiveness and lower sufficiency are preferred.

Observed trade-off: compared descriptively with our Phase 1 LIME row, WA-SHAP has higher comprehensiveness in Action (0.659), Object (0.673), and Location (0.508). Its sufficiency is higher in all three slots, which is worse under this metric. Therefore, the correct claim is not that SHAP is superior on every faithfulness measure.

Configuration caveat: the Phase 1 baseline and the Phase 2 SHAP run do not have perfectly matched alignment and feature-definition details. This full-test comparison is supporting evidence; the controlled stability study on the next page is the main test of the Phase 2 improvement.

Limitations and honest interpretation

1. WA-SHAP is exact only for the 83.34% of short utterances whose coalition spaces fit the budget; it remains a sampled approximation for longer commands.

2. The matched stability experiment favors WA-SHAP. Full-test faithfulness is not feature-matched: its sufficiency was worse than saved Phase 1 LIME results. No classifier-accuracy improvement is claimed.

3. The full-test Phase 1 versus Phase 2 faithfulness rows have an alignment/feature-definition caveat. The matched word-only stability comparison is therefore the strongest causal comparison.

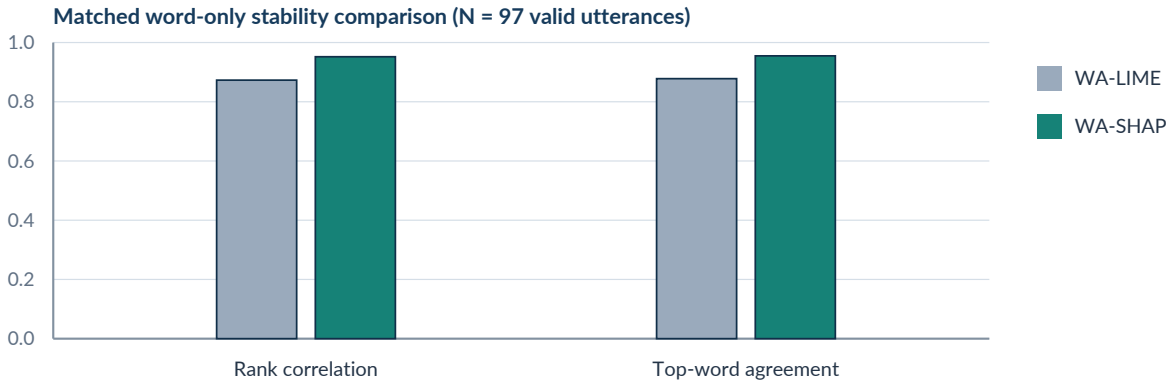
4. Noise-matched masking is a controlled alternative to digital silence, not a guarantee of in-distribution or natural speech audio.

4. Controlled experimental evaluation: stability across seeds

To remove the empty-span confound, LIME was rerun using the same word-only features as WA-SHAP. Both methods were evaluated on the same 100 sampled FSC utterances using five seeds (42, 43, 44, 45, 46). Three degenerate examples had undefined rank/top-word statistics, leaving 97 valid clusters.

Method	Valid utterances	Mean Spearman rank correlation	Top-word agreement
WA-LIME (controlled, word-only)	97	0.873 +/- 0.018	87.8% +/- 2.7%
WA-SHAP (same valid subset)	97	0.952 +/- 0.022	95.1% +/- 2.6%

Table 3. Our implementation: cluster-level means with approximate 95% confidence-interval half-widths. Higher is better for both stability measures.



Interpretation: after matching the word features, WA-SHAP remains more stable than WA-LIME. Its mean rank correlation rises from 0.873 to 0.952 and its top-word agreement rises from 87.8% to 95.1%. This is the primary evidence that the proposed method addresses LIME's sampling-instability limitation.

5. Secondary study: masking sensitivity

Gap 1: masking sensitivity

Explainer	Slot	Compr. delta: noise - silence	Suff. delta: noise - silence
LIME	action	-0.049	+0.057
LIME	location	-0.081	+0.035
LIME	object	-0.027	+0.024
LOO	action	-0.017	+0.031
LOO	location	-0.077	+0.023
LOO	object	-0.009	+0.009

Table 4. Our implementation: Gap 1 comparison under a fixed hard-silence evaluator. Noise-matched attribution gives lower comprehensiveness and higher sufficiency in all listed cases. This demonstrates evaluator-mask coupling; it does not establish that synthetic noise is inherently worse.

Conclusion

The project successfully implements and evaluates WA-SHAP as a Phase 2 extension of word-level speech explanations. The controlled experiment supports the claim that WA-SHAP is more stable than WA-LIME under matched word-only features. Faithfulness results show stronger comprehensiveness together with a sufficiency trade-off, which is reported transparently.

References

1. Pastor, E., Koudounas, A., Attanasio, G., Hovy, D., and Baralis, E. (2024). Explaining Speech Classification Models via Word-Level Audio Segments and Paralinguistic Features. *Proceedings of EACL 2024*, pp. 2221-2238.

2. Lundberg, S. M., and Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems 30*. [Supporting article for KernelSHAP.]

3. Covert, I., Lundberg, S., and Lee, S.-I. (2021). Explaining by Removing: A Unified Framework for Model Explanations. *Journal of Machine Learning Research*, 22(209), 1-90.

4. DeYoung, J., et al. (2020). ERASER: A Benchmark to Evaluate Rationalized NLP Models. *ACL 2020*. [Source of the comprehensiveness and sufficiency framing used by the base paper.]